

Suman Janarthanan

Github | LinkedIn | Portfolio
surplussuman987@gmail.com | 6383595092 | Dharmapuri, Tamil Nadu

EDUCATION

IIT GUWAHATI

DATA SCIENCE (CREDIT PROGRAM)
Mar 2026 | Grade: 70%

I.K.G. PTU, PUNJAB

B. TECH IN COMPUTER SCIENCE
Jun 2025 | CGPA: 7.71/10

GOVT MODEL HR SEC

Grad. April 2020 | JEE: 80%ile

SKILLS

AI ARCHITECTURE

Orchestration & Design:

RAG Pipelines • LangChain / LangGraph •
Multi-Agent Orchestration • Hybrid
Model Routing • Vector Databases •
Fine-Tuning Pipelines

Core & Libraries:

Python • PyTorch • TensorFlow •
OpenCV • Computer Vision • NLP

CLOUD & DEPLOYMENT

AWS Architecture:

EC2 • S3 • WAF • IAM • Security Groups
/ NSGs • ALB/NLB • VPC Design • Route
53 • Auto Scaling • CloudWatch

DevOps:

Zero-Downtime CI/CD • Docker • Nginx
• Unicorn • Prometheus / Grafana

BACKEND STACK

Django REST • FastAPI • Redis • Celery •
PostgreSQL • MongoDB • WebSockets /
WebRTC

EXPERIENCE

T.I.M.E (TRIUMPHANT INSTITUTE OF MANAGEMENT

EDUCATION) | TECHNICAL LEAD --- AI ARCHITECT & CLOUD
INFRASTRUCTURE

Feb 2026 -- Present | Hyderabad

- Own end-to-end **AWS deployment architecture** — WAF, IAM, Security Groups & NSGs, VPC design, and Load Balancers — with **zero-downtime CI/CD** and Grafana/Prometheus observability, replacing a provider plagued by crashes and outages.
- Architected **ARIA**, an adaptive AI mentor built on continuous behavioral profiling, and an AI-proctored interview platform combining live behavior analysis with adaptive scoring, delivering **5,000+ interviews**.
- Designed **Lexio**, an 8-stage communication assessment platform, and a dual-agent verification pipeline converting 150+ page legacy documents with zero content loss.

DMX TECH SERVICES PVT LTD | AI ENGINEER & TECHNICAL TEAM LEAD

Feb 2025 -- Jan 2026 | Bangalore

- Led 10+ cross-functional teams (10-25 engineers) delivering production AI systems serving **15,000+ active users** at **99.7% uptime**.
- Architected a hybrid multi-model routing strategy cutting operational costs **85%** while improving output quality by **23.6%**.
- Designed CI/CD pipelines and LLD standards, cutting production bugs **65%** and lifting team productivity **40%**.

SIRUSAI PRIVATE LIMITED | AI DEVELOPER

Aug 2023 -- Oct 2024 | Dharmapuri

- Engineered a surveillance facial recognition system achieving **95% accuracy** using WebRTC for real-time P2P CCTV streaming, deployed via Docker on VPS infrastructure.

KEY PRODUCTION PROJECTS

ARIA: ADAPTIVE RESOURCE INTELLIGENCE ANALYST | CONTINUOUS BEHAVIORAL PROFILING + RAG

- Fused test performance, video-watch behavior, and doubt-asking patterns into one longitudinal student profile that auto-detects **recurring mistake patterns** instead of isolated test scores.
- Delivered a **continuously updated** improvement plan, not a static report, deployed as the primary AI mentor across the institution's learning platform.

AI MOCK INTERVIEW & PROCTORING SUITE | BEHAVIORAL CV + ADAPTIVE SCORING ARCHITECTURE

- Layered **live webcam behavioral analysis** (engagement, nervousness signals) on top of response scoring to judge interview readiness holistically, not just answer correctness.
- Delivered **5,000+ interviews** with longitudinal scoring that tracks candidate improvement across sessions, run at institutional scale for thousands of concurrent mock interviews.

PHILOSOPHY

PRODUCTION-FIRST

Research is exciting, but real users need reliability. I build for 99.7% uptime.

OWN THE FULL STACK

I design the AI system and the AWS infrastructure it runs on — the full path, not just the boundary.

AWARDS

Nov 24 SIH (ISRO PS)
Sep 23 1st State Ideathon
May 23 2nd Udhyam Govt
Apr 23 1st Best Innovation
Mar 23 1st Prog2Entrepreneur

PUBLICATIONS

Neuralink: Advancing BMI for Enhanced Human-Machine Interaction | Link (2024)

CPU-Efficient Face Recognition for Low-Light, Long-Range Surveillance | In Preparation (2026)

LEXIO: COMMUNICATION IMPROVEMENT PLATFORM |

8-STAGE DIAGNOSTIC ARCHITECTURE

- Designed an **8-level diagnostic engine** (Speaking, Writing, Vocabulary, Presentation, Grammar, Reading, Listening, Group Discussion) independently scoring every dimension of communication.
- Built an **AI-simulated group discussion** module to assess collaborative communication, not just solo responses.
- Generated a personalized study plan, material, and task list directly from each learner's diagnostic profile.

CLOUD CODING & COMPILER PLATFORM | DOCKER,

WEBSOCKETS, HORIZONTAL SCALING

- Re-engineered infrastructure from 100 to **5,000+ concurrent users** (**50x** capacity) while cutting cost **58.5%**.
- Achieved **95.7% faster** response (4.2s to **180ms p95**) via connection pooling and worker-count tuning, sustaining **99.8% uptime**.
- Built secure, sandboxed multi-language execution (20 languages) supporting **100+ live coding contests** with zero crashes.

STUDY MATERIAL BEAUTIFICATION PIPELINE | DUAL-AGENT

VERIFICATION, CHUNKED BATCH PROCESSING

- Designed a conversion-plus-review agent pair turning **150+ page** legacy documents into HTML with **zero content loss**, chunked to fit context limits and cut cost.

SMART EXAM EVALUATION SYSTEM | HYBRID OMR/OCR PIPELINE

- Automated grading for **160+ centers**, processing **600,000+ answer sheets** in 8 months at **98.7% accuracy**, cutting evaluation cycles from 15 days to 18 hours.

INTELLIGENT DOCUMENT ANALYSIS | OCR + RAG VERIFICATION

PIPELINE

- Raised throughput **140x** (500 docs/week to 10,000 docs/day) with **96.8%** fraud detection across 8 universities.